

Tax Administration and Firm Performance: Evidence from VAT Refund Delays *

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Abstract

This paper shows how weak tax administration leads to economic costs for firms in developing countries. We study the case of VAT refund processing in Zambia, where, as we document in a novel dataset, firms wait 700 days for reimbursement on average. Linking our dataset to the universe of tax returns at the firm level, we establish several stylized facts on the pitfalls of refund processing and estimate the economic costs of refund delays for firms. To do so, we rely on an instrumental-variable approach which exploits plausibly exogenous variation in administrative congestion at the time of claiming. The results show that prolonged refund delays significantly reduce firms' sales, taxable purchases, profits, investment, and employment, consistent with firms facing binding working-capital constraints while awaiting reimbursement. These findings suggest that the effectiveness of tax systems depends not only on statutory design but also on the capacity of tax administrations to implement them.

JEL codes: H25, H26, O23.

Key-words: Value-Added Tax, Tax Refunds, Zambia, Tax Compliance

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1 Introduction

Policy outcomes depend fundamentally on implementation, as even well-designed recommendations hinge on administrative execution. Beyond formal laws and statutory rates, this administrative reality defines the regulatory framework in which firms operate. In taxation, state capacity to execute policy can therefore have real economic consequences that extend far beyond formal tax rules.

One area where such concerns are widespread lies in administering the value-added tax (VAT), which has become a cornerstone of modern tax systems (Brockmeyer et al., 2024). Its theoretical advantages are well established: it is difficult to evade, generates substantial revenue, and does not distort firms’ production decisions. These properties are especially attractive in lower-income countries, where state capacity is low and tax revenues but also economic development are direly needed. Yet low state capacity, which makes the VAT appealing in theory, can undermine its efficacy in practice. In other words, the circumstances that make a well-functioning VAT desirable are precisely the circumstances under which administering one is hard.

A particular pressure point in the functioning of the VAT is the timely payout of tax refunds, which establish production-efficiency. Such payments are due when a firm has paid more tax than it actually owes. For example, when the tax it paid on inputs exceeds the tax it is supposed to pay on outputs, i.e., its taxable value added is negative. Delays in this process impose liquidity constraints on firms, erode trust in the system, and can ultimately undermine the very benefits that motivated adopting the VAT in the first place. While the refund system has long been acknowledged as a key administrative bottleneck (Ebrill et al., 2001) and there is no lack of anecdotal evidence that developing countries struggle with it (Harrison and Krelove, 2005), surprisingly little is known about the actual consequences for real economic outcomes.

In this paper, we quantify the real costs that VAT refund delays impose on firms. Our empirical setting is Zambia, where the malfunctioning of VAT administration has led to such discontent among firms that it was nearly abandoned in 2021. For our analysis, we construct a novel dataset on the universe of VAT refund claims and payout dates. From the microdata, we establish stylized facts which illuminate the to-date black box of refund administration. We document that refunds are a widespread phenomenon affecting the whole economy and that the amount of refund claims have outpaced the amount of refund payments by far. This divergence was

exacerbated by the expansion of Zambia’s withholding scheme, which pushed many firms into a refund position with the government, highlighting the importance of balancing different policy goals. Among claims that are settled by the end of our study period in 2023, we document severe delays of about 700 days on average, reflecting a delay of more than 2300 %. We further highlight that the type of refund settlement has changed over time, with the majority of payments shifting from direct transfers to offsetting future liabilities, with potential implications for the erosion of other tax bases.

We propose a simple theoretical framework to guide our empirical analysis. A firm chooses output and input to maximize profits, pays VAT on both but is liable for a refund if the latter tax exceeds the former. In the absence of refund delays, production efficiency is restored. With positive delays, the effective price of inputs rises, leading the firm to cut capital and output below the levels it would otherwise choose. The model produces two further predictions. First, distortions grow with delay length and, second, distortions are larger for firms whose withheld refund is large relative to their input base.

To test these predictions and estimate the real effects of refund delays, we merge the refund dataset with the universe of firms’ VAT and Corporate Income Tax (CIT) filings, allowing us to follow firms before, during and after refund episodes. Because refund delays are potentially endogenous to firm characteristics, we exploit plausibly exogenous variation arising from fluctuations in the Zambia Revenue Authority’s (ZRA) administrative processing capacity. Specifically, we instrument each firm’s refund delay with the leave-one-out mean delay of other firms entering the refund queue in the same quarter (3 months). The instrument is relevant because cohort-wide congestion strongly predicts individual firm delays. It arguably satisfies the exclusion restriction because, controlling for sector and size, ZRA’s processing backlog is plausibly determined by administrative capacity, not by the characteristics or outcomes of any individual claimant. We rule out that our instrument merely picks up selection or general macroeconomic effects by running a series of checks, including pre-trends analyses, placebos and testing whether congestion has any effect on firms which never claim for refunds.

Our baseline results show find that longer refund delays substantially reduce firms’ sales, purchases, profits, and capital. The effects are economically large, statistically significant and consistent across specifications. Importantly, we verify the headline estimates using both VAT and CIT information on the same outcome. The results suggest that a delay of additional quarter, induced by our measure of administrative congestion, reduces a firm’s sales by about 4.5 – 5.4 % relative to the

time before refunds were claimed. We find similar magnitudes for purchases. These effects are remarkably similar across sectors, confirming that refund delays affect the whole economy. Dissecting the effects by refund intensity, we find that the results are driven mainly by firms for whom the withheld refund represents a large share of working capital, consistent with our theoretical prediction.

We confirm our results in several robustness checks. One potential concern is that the baseline outcome averages firm performance over the entire delay window, which mechanically weights longer-delay firms more heavily. We address this by measuring the outcome over fixed windows instead, removing any correlation between window length and the outcome mean. The results remain consistent and confirm that the effects worsen with longer wait time. We also show that results are stable across specifications that absorb sector, province, and sector-by-entry-year fixed effects, and that the findings are consistent across the two independent data sources (VAT and CIT records) which measure overlapping but distinct sets of firm outcomes. It is important to note that the majority of refunds which have been claimed throughout our study period are still outstanding. This means that our estimation sample is biased toward firms with shorter delays but we show that excluding firms whose refunds have not been settled by the end of our study period is unlikely to drive our results. Taken together, the results suggest that the capacity of the tax administration has a direct effect on economic performance beyond statutory rates or design.

This paper contributes to several strands of the literature. First, we add to the literature on how the institutional context of developing countries shapes public administration (e.g., [Gordon and Li, 2009](#); [Besley and Persson, 2013](#)). Most of this work takes a broad view of state capacity. By the virtue of our new dataset, we are able to zoom to the firm and tax-return level, showing how a specific administrative failure translates into real economic distortions. Whilst suspected by practitioners and researchers for a long time, we show that the concern about refunds is borne in the micro-data. Our findings therefore deliver important novel evidence that delays in VAT refunds do harm firms.

Second, we contribute to the growing literature on VAT systems in lower-income countries, which has highlighted that in these setting both taxpayers as well as the state may be unable to properly enforce or comply with it ([Brockmeyer et al., 2024](#); [Keen and Lockwood, 2010](#)). For example, [Almunia et al. \(2024\)](#) document filing inconsistencies in Uganda, which can partly be explained by mistakes rather than strategic evasion. [Waseem \(2023\)](#) studies the VAT in Pakistan and provides evidence that firms are misusing the refund system to create spurious claims, so called

invoice mills. Most closely related to our work are studies by [Pinto et al. \(2024\)](#) and [Brusco et al. \(2024\)](#) who study how refund policy affects firms in Honduras and South Africa respectively.¹ The former study a specific and indirect kind of refund reform, namely the reduction of withholding taxes of digital providers, which led to a decrease in effective tax rates for firms who held large refund claims against the government. Despite this decrease, they cleanly rule out effects on real firm outcomes. Our findings focus on delays in refunds directly and are not inconsistent with these results from Honduras. While we document a working capital constraint during long waiting periods, the authors essentially document the absence of a rebound effect once the claims are cleared. While not using information on individual refund claims, [Brusco et al. \(2024\)](#) document that, on average, firms invest more in capital after the tax administration increased the speed of audits and thereby reduced the delay of refunds in South Africa. Albeit complementary to our findings, the magnitude of delays considered is negligible compared to the Zambian case we document in this paper.² To the best of our knowledge, our paper complements these studies by providing the first micro-evidence that longer refund delays reduce economic performance. Additionally, we document this in a low-income setting³, where administrative constraints and firm vulnerability are most severe.

Third, we contribute to the broader literature on the real economic effects of public administration (e.g., [Basri et al., 2021](#); [Best et al., 2023](#); [Aman-Rana and Minaudier, 2026](#)). Within this realm, we show how de-facto deviations from first-best official processes undermines the very benefits of well-intended policies.

The remainder of this paper proceeds as follows. Section 2 discusses the institutional background and Section 3 describes the data and establishes stylized facts on Zambia’s refund system. Section 4 develops a simple theoretical framework and Section 5 discusses the empirical strategy. Section 6 reports our results. Section 7 concludes.

2 Background

Zambia is a lower middle-income country and classified as a least developed country according to the [United Nations \(2023\)](#). In 2021, it had a population of 20 million

¹[Chandra and Long \(2013\)](#) also studies how tax rebate delays affect firms, though in China, which differs substantially from low-income Sub-Saharan Africa.

²[Brusco et al. \(2024\)](#) use a policy change which reduced the average waiting times from only 30 days to 10 days.

³Zambia moved from low-income to lower-middle-income status only in 2023 according to World Bank classification, having switched back and forth in prior years.

people, a GDP-per-capita of USD 1137 PPP and a tax-to-GDP ratio of about 16%. It thus closely resembles the average Sub-Saharan African country along these dimensions ([ATAF, 2018](#)). In principle, every business in Zambia must be registered with the tax authority (ZRA). However, as in all lower income countries, most firms in Zambia are small and informal (not registered with the tax authority). Official statistics estimate that the informal sector accounted for nearly 90% of the country’s employment in 2014 ([Ministry of Labour and Social Security, 2018](#)). Among the firms that are under the tax net, there is a crucial size distinction for how the tax base is determined. Businesses with annual turnover above ZMK 800,000 ($\approx$ USD 31,000) are liable for the corporate income tax (CIT) where taxes apply to profits. Additionally, these businesses are required to register for the VAT.⁴

There have been numerous debates on whether Zambia should continue its implementation of a value-added tax instead of a sales tax since its inception in July 1995. However, with its initial rate of 17.5%, the country has maintained its tax policy by altering rates in previous years to now 16% since 2008. At this rate, the tax contributes close to a quarter of domestic revenues ([ZRA, 2021](#)).

2.1 Refund administration

VAT refunds are a key part of the VAT system. If a firm’s input tax exceeds its output tax, the firm has in fact paid too much taxes and thus the government owes the firm and has to pay out a refund. In Zambia, VAT revenues make up a substantial share of the government’s revenue and are thus of prime importance to policy makers. Therefore, VAT payments are transferred from the ZRA to the Ministry of Finance immediately. Only a small share is held back by ZRA for cases when refunds need to be paid out. Given the strong delays in refund payments, this amount was clearly miscalculated. Additionally, the total amount allocated to refund purposes is inflexible. Each month, the total amount of refunds could therefore not exceed a certain amount, irrespective of the demand for refunds or the state of the economy. The amount was capped at ZMK 800 mio up until 2021. With the reform in November 2021, it nearly doubled to ZMK 1.35 bio.

Refund claims in Zambia are processed centrally by dedicated refund officers within the Domestic Taxes Division of the Zambia Revenue Authority (ZRA). Once a claim is submitted, it enters a processing queue that is largely determined by the date of submission and administrative procedures such as risk assessment, document

⁴Voluntary registration for VAT is possible but very low in Zambia, potentially due to the malfunctioning of the refund system we document in this paper.

verification and audit requirements. Individual firms have little ability to influence their position in this queue after filing. Consequently, periods with unusually high volumes of refund claims generate administrative congestion and backlogs that affect all firms entering the refund system at the same time. Since the processing capacity available to a particular cohort of refund claimants is determined by aggregate administrative conditions rather than firm-specific characteristics, firms are effectively exposed to different states of administrative capacity depending on when they happen to submit their refund claims. Firms do not receive interest on the withheld refunds such that refund delays effectively act as interest-free loans to the government.

3 Data

In this section, we discuss the different data sources. We begin by describing the refund claim and payment data before moving to the data on the actual tax returns.

3.1 VAT refund data

The main component to our analysis is a novel dataset on refund settlements. This dataset is the result of a large collection effort in collaboration with the ZRA, including the merging of different, partly manual sources. It contains information on all refund payments made by ZRA between 2014 and 2023. Along with the tax ID and the paid amount, the records include the actual payout date as well as the original claim date. Via this claim date and the tax ID, we are able to match the these payments directly to the original claims. All submitted claims go through an audit process known as a verification check. The data states whether and by how much the initially claimed amount has been adjusted before it was approved.

There are two ways how refund claims are being settled. First, there can be outright payments, i.e. taxpayers receive their refunds in cash or as a bank transfer. Second, there can be offsets, i.e. taxpayers are allowed to use the approved refund as an offset against other tax liabilities. In some cases, the total refund amount is divided into an offset amount as well as an outright payment amount. Overall, we have information on 81,350 refund settlements, with the oldest claims dating back to 1995. It is important to note that these data only include claims that have been settled and that they do not capture refunds which were still outstanding at the end of 2023.

3.2 Tax return data

To investigate the relationship between refunds and firm outcomes, we further use administrative data from VAT and CIT returns. The former allows us to track monthly outcomes such sales and expenditures, the latter reports annual measures such as the value of firm income as well as other balance sheet information.

The VAT data covers a filer population of almost 85,000 firms over the 10-year period we study. The panel contains information on all line items available in the VAT return (Overbeck et al., 2026). Importantly, it provides a benchmark and proof of eligibility of a firm’s refund claim. This is an essential complement as it allows tracking refund claims which have not been (at least partially) settled by 2023 and therefore do not show up in the refund payment data. VAT filing is required monthly but in practice there are filing gaps at the month level. These can largely be explained by unstable monthly gross output of firms who consequently do not file. They often compensate for this inconsistency by reporting the quarterly aggregate each quarter. Gaps in monthly VAT filing therefore do not necessarily signal evasion. For these reasons, our analysis aggregates all VAT variables at the quarterly level. While this approach discards any within-quarter dynamics, we consider it a valid trade-off to smoothen inconsistencies. Furthermore, as refund delays usually exceed 1 month by far, we believe that little information is lost due to aggregating months to quarters.

The CIT data comprise the main components of the balance sheet and income statement such as gross profits, assets (debts, liability, investment) or employment. We focus on firm performance variables that correlate to liquidity injection resulting from tax refunds. In addition to these operational information, the data contain demographic indicators such as location and industry.

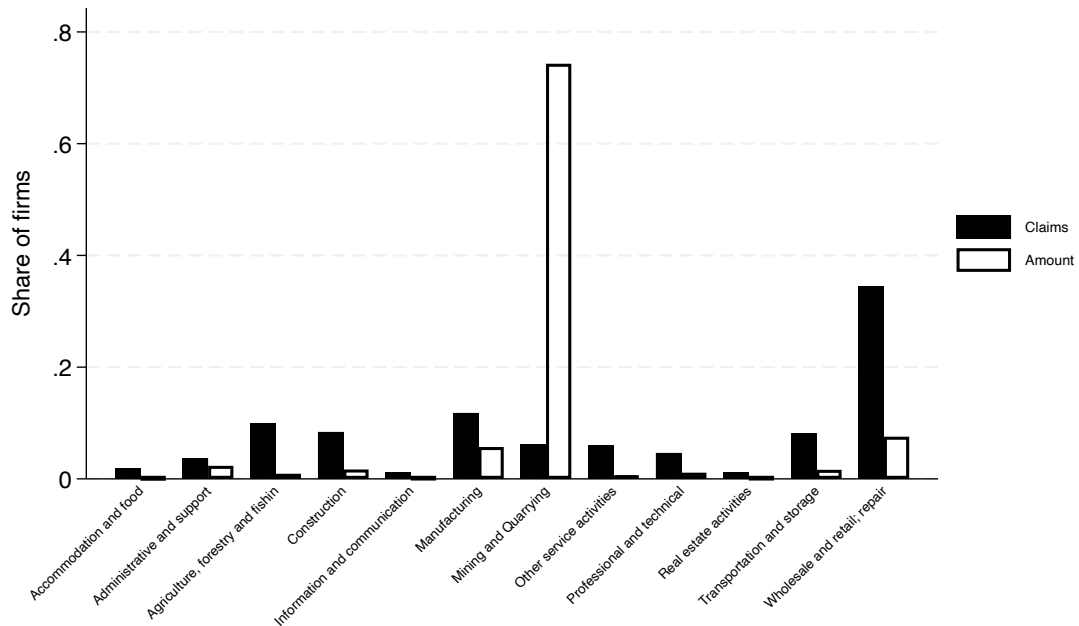
3.3 Descriptive statistics

We establish four stylized facts on Zambia’s VAT refund administration.

Refunds are economy-wide. Refunds are not constrained to a small number of large firms but affect the whole economy. However, a distinction has to be made between the amount of claims that are made and the actual amount of money which is claimed. As shown in Figure 1, refund claims (black bars) are spread across nearly all main sectors. Wholesalers and Retailers take the lion share of about 34%. Manufacturers claim about 12% of refunds, mining 6% and agricultural firms claim 10% of

refunds. This insight is important as it shows that not only large influential mining firms are affected by the refund system, as often claimed by observers. Turning to the amounts which are being claimed, however, mining firms are in fact leading with 74%. Wholesalers only make up 7.5% of the total amount. It is to note that these figures are absolute numbers and do not permit conclusions about relative importance of refunds within sectors. The imbalance in the claimed amounts is partly to be expected as Zambia’s exports, which are more prone to trigger refunds, are heavily concentrated on the mining sector.

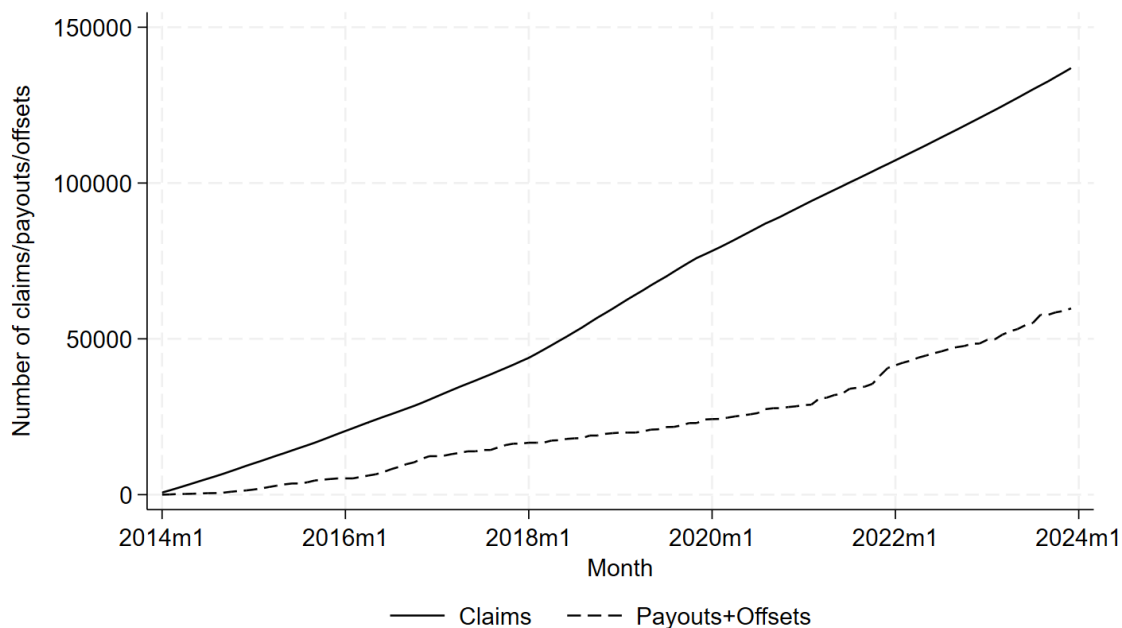
Figure 1: Distribution of VAT refunds across sectors



Notes: This figure plots distribution of VAT refund claims and VAT refund claim amounts respectively over different industries. Data source: ZRA. Years: 2014-2023.

Refund claims outpace settlements – by far. Figure 2 plots the cumulative evolution of claims filed since 2014 and payments or offsets made since 2014 in respect of prior claims. Two patterns stand out. First, there is a strong and persistent divergence between the two lines: claims grow substantially faster than payments, so the share of outstanding claims grows over time. Second, the pace of divergence accelerates in 2018, with a clear kink in the claims series while the payments series remains stable. This kink coincides with a major expansion of the Withholding VAT Regime, under which designated agents began to withhold output VAT from a larger number of suppliers (Adu-Ababio et al., 2023). Withholding mechanically

Figure 2: Divergence of claims and refunds



Notes: This figure plots the cumulative sum of VAT refund claims and VAT refund payments per year. Data source: ZRA. Years: 2014–2023.

shifts firms' VAT positions: suppliers that previously remitted net VAT now accumulate excess input credits because the VAT they would have collected from customers is withheld at source. As these firms continue purchasing inputs on which they pay VAT while receiving less output VAT, persistent refund positions emerge.⁵ Overall, Figure 2 shows that the system is incapable of servicing the volume of refunds being claimed, necessarily generating substantial delays.

Delays are severe. For each settled refund claim we calculate the time between claim and first settlement, compared to the legal deadline of one month. Appendix Table A.1 details the distribution of delays. On average, it takes more than 22 months longer than legally required for a refund to be paid out, implying an average delay of 2200%. About 10% of refunds take more than four years; 5% take more

⁵This is particularly salient for Wholesale & Retailers who often supply to larger withholding firms, and the data show a roughly 50% increase in monthly claims from this sector between December 2017 and April 2018. The combination of more firms in a refund position and greater documentation requirements for each claim necessarily increased the workload of refund-processing staff, culminating in an overburdening of the system. While the timing and sectoral pattern of the surge are fully consistent with this mechanism, the expansion affected all eligible firms in the relevant sectors simultaneously. This leaves no within-sector control group and precludes the firm-level variation needed for causal identification; we therefore consider it an interesting institutional feature rather than a source of identifying variation.

than 5.5 years. Given that roughly a third of Zambian firms do not operate for longer than ten years (WB, 2019), many firms thus likely never receive the refunds they are owed. It is to note that these figures only take into account refunds that have been settled and thus underestimate the true extent of delays.

There is substantial heterogeneity across sectors. Mining firms wait on average 14.4 months, roughly half the waiting time of Wholesale & Retail firms (25.6 months) and one third below the overall average. This finding is consistent with the notion that mining firms have established a reputation as reliable taxpayers coupled with ZRA deliberately prioritizing large claims and therefore are able to capture refunds faster. Manufacturing firms face 50% longer delays than mining firms on average confirming perceptions of unequal treatment by ZRA and frustration in the sector (Financial Insight Africa, 2024). We argue that these delays are in fact due to slow administrative processing instead of a mere shortage of resources for two reasons. First, delays are slightly smaller but comparable for firms which receive offsets, which de-facto do not require liquidity of the government. Second, Appendix Figure A.1 shows that the total monthly payments fluctuate strongly around statutory amount dedicated to refund payments (cf. Section 2) and even exceed these consistently after 2021.

More offsets, less payouts. We document a remarkable shift in refund policy from cash or bank transfers as settlement form to offsets, that is, crediting the taxpayer’s refund against present or future tax obligations. Appendix Figure A.1 documents this: while nearly all refunds were settled in outright payments before 2021, from 2021 onwards total refund amounts increased substantially but the share paid directly fell sharply. Mining firms provide the starkest example: from 2021, their claims are settled exclusively via offsets against other tax liabilities.

This shift has two potential implications. First, for our empirical analysis, it means that refund recipients received a mix of cash and offset settlements. The real-effects estimates therefore capture the combined liquidity value of both forms of settlement, which may differ if firms face binding constraints on using offset credits. Second, from a policy perspective, systematic reliance on offsets could carry risks for the broader tax base. If overstating VAT-creditable purchases is administratively easier than evading the CIT, an offset system inadvertently creates incentives to channel evasion through the VAT (Waseem, 2023), eroding CIT collections and undermining the credibility of the refund mechanism. Designing sound refund policy therefore requires balancing the short-term fiscal advantages of offsets against these longer-run compliance risks.

4 Theoretical Framework

In this section, we provide a simple theoretical framework on how VAT refund delays affect firm decisions. We consider a firm which produces a final product according to a quasi-concave production function $F(\cdot)$. There is only one input good k , which can be purchased at price p . VAT is modeled as a tax on outputs τ_O and an input tax τ_I that applies on input purchases. For simplicity, we assume complete pass-through such that both taxes are simply added to the final price. If the input taxes exceed output taxes, the firm receives a refund. To incorporate this possibility of refunds, we allow τ_O and τ_I to differ. We model the possibility of delayed refunds by letting $\rho(d) \in [0, 1]$ denote the effective value to the firm of a refund paid with delay $d \geq 0$, where $\rho(0) = 1$ and $\rho'(d) \leq 0$. The parameter $\rho(d)$ may reflect the time value of waiting or, more relevant in settings with imperfect capital markets, the firm's inability to borrow against an outstanding refund claim. The after-tax profits of the firm $\Pi(k)$ can be written as:

$$\begin{aligned} \Pi(k) = & F(k)(1 + \tau_O) - kp(1 + \tau_I) \\ & - \underbrace{\max[(F(k)\tau_O - kp\tau_I), 0]}_{\text{tax payment}} \\ & + \underbrace{\max[(kp\tau_I - F(k)\tau_O), 0]}_{\text{refund}} \times \rho(d) \end{aligned} \quad (1)$$

where $\rho(d) = 1$ corresponds to an immediate refund and $\rho(d) = 0$ to no refund being paid. The firm chooses input purchases optimally in order to maximize profits. Its objective is given by

$$\max_k \Pi(k). \quad (2)$$

Given this objective it is worthwhile to consider three different scenarios:

Case 1: $kp\tau_I < F(k)\tau_O$:

There are no refunds due and the first order condition (FOC) reads

$$F'(k) = p \quad (3)$$

which is the same as in the absence of any taxes. In this case, the VAT reveals its production-efficient property.

Case 2: $kp\tau_I > F(k)\tau_O$ and $\rho(d) = 1$:

The tax payment is zero but the refund is positive and paid immediately. As before,

the VAT remains production efficient and the FOC is as in (3).

Case 3: $kp\tau_I > F(k)\tau_O$ and $\rho(d) \in [0, 1)$:

In this case, the firm is liable for a refund but receives it only partially or with delay. Rearranging the profit function yields

$$\Pi(k) = F(k)[1 + \tau_O(1 - \rho(d))] - kp[1 + \tau_I(1 - \rho(d))], \quad (4)$$

and the FOC reads:

$$F'(k) = p \frac{1 + \tau_I(1 - \rho(d))}{1 + \tau_O(1 - \rho(d))}. \quad (5)$$

The firm's decision is distorted by the factor $\frac{1 + \tau_I(1 - \rho(d))}{1 + \tau_O(1 - \rho(d))}$, which equals one when $\rho(d) = 1$ and approaches $\frac{1 + \tau_I}{1 + \tau_O}$ as $\rho(d) \rightarrow 0$.

Lemma 1: *If the firm is profitable (before taxes) and is liable for a refund it needs to hold that $\tau_I > \tau_O$.*

Proof: For the sake of contradiction, assume that $\tau_I < \tau_O$. As the firm is not making losses it holds that $F(k) > kp$. This implies that $F(k)\tau_O > kp\tau_O > kp\tau_I \Rightarrow F(k)\tau_O > kp\tau_I$, which contradicts that a refund is due. $\square$

By Lemma 1, $\tau_I > \tau_O$, and therefore the distortion factor in (5) exceeds one for any $\rho(d) < 1$. Moreover, since $\frac{1 + \tau_I(1 - \rho)}{1 + \tau_O(1 - \rho)}$ is decreasing in $\rho(d)$ and $\rho'(d) \leq 0$, the distortion is *increasing in delay d*: a longer delay reduces the effective value of the refund, raising the implicit cost of inputs and depressing optimal input demand below its efficient level. Our framework thus predicts that input purchases and output fall monotonically in the length of the delay, with full efficiency restored only when $\rho(d) = 1$.

The optimality condition in Eq. (5) further implies that the magnitude of the distortion may vary even across firms facing the same delay. To see why, note that by Lemma 1, $\tau_I > \tau_O$, so the right-hand side exceeds p for any $\rho(d) < 1$. The size of the distortion depends on the difference $\tau_I - \tau_O$: a firm that faces a high input tax rate relative to its output tax rate suffers a larger wedge than one where the two rates are close, even if both wait the same number of quarters for their refund.

To arrive at a clear relationship between this wedge and efficiency it is useful to log-linearize Eq. (5) around the benchmark where $\ln F'(k_0) = \ln p$:

$$\ln F'(k^*) - \ln F'(k_0) = \ln[1 + \tau_I(1 - \rho(d))] - \ln[1 + \tau_O(1 - \rho(d))]. \quad (6)$$

Using $\ln(1+x) \approx x$ for small x , the right-hand side can be approximated by

$$\ln \left[\frac{1 + \tau_I(1 - \rho)}{1 + \tau_O(1 - \rho)} \right] \approx (\tau_I - \tau_O)(1 - \rho(d)).$$

Considering the simple Cobb-Douglas example of $F(k) = Ak^\alpha$, the left-hand side of Eq. (6) reduces to $(\alpha - 1) \Delta \ln k$.⁶ Thus, the equation reduces to:

$$\Delta \ln k \approx \frac{(\tau_I - \tau_O)(1 - \rho(d))}{\alpha - 1}. \quad (7)$$

Since $\alpha < 1$, the denominator is negative, and by Lemma 1 the numerator is positive for any $\rho(d) < 1$, so $\Delta \ln k < 0$. This implies that delays unambiguously reduce optimal input demand. Two further implications follow from Equation (7). First, the distortion is increasing in delay d : since $\rho'(d) \leq 0$, a longer delay raises $1 - \rho(d)$ and therefore the magnitude of $\Delta \ln k$. Second, the distortion is increasing in the tax wedge $\tau_I - \tau_O$: conditional on the same delay, firms with a larger gap between input and output tax rates face a larger reduction in optimal inputs.

In our Cobb-Douglas example, applying $\Delta \ln y = \alpha \cdot \Delta \ln k$ directly translates the input distortion into an output loss:

$$\Delta \ln y \approx \frac{\alpha (\tau_I - \tau_O)(1 - \rho(d))}{\alpha - 1}. \quad (8)$$

The sign is again negative, implying that longer waiting times as well as a wider tax wedge both independently reduce output.

5 Empirical Strategy

Identifying the causal effect of refund delays on firm outcomes is complicated by the fact that delays may be endogenous to other characteristics of the firms which independently affect our outcomes of interest. Firms that file large or persistent refund claims tend to be larger and more export-oriented, with stronger underlying growth prospects that attenuate or reverse any estimated negative effect in OLS. Also, anecdotal evidence suggests that firms with better political connections face systematically shorter delays. We address this by exploiting plausibly exogenous

⁶For a general concave F , the left-hand side of Eq. (6) is approximated by a first-order Taylor expansion of $\ln F'(k)$ in k : $\ln F'(k^*) - \ln F'(k_0) \approx \frac{d \ln F'}{dk} \Big|_{k_0} \cdot \Delta k = \frac{d \ln F'}{d \ln k} \Big|_{k_0} \cdot \frac{\Delta k}{k} \approx \frac{d \ln F'}{d \ln k} \Big|_{k_0} \cdot \Delta \ln k$, where the last step uses $\Delta k/k \approx \Delta \ln k$. Under Cobb-Douglas, $\ln F'$ is linear in $\ln k$, so both approximations are exact and $\frac{d \ln F'}{d \ln k} = \alpha - 1$ everywhere.

variation in administrative congestion using an instrumental variable (IV) design.

Defining delays. As many firms claim refunds more than once, and claims can overlap it is difficult to think of the effect that a single delayed refund claim has on a firm. Therefore, to still arrive at a single firm-level measure of delay, we track each firm’s *outstanding balance*, that is, the cumulative sum of approved refunds less cumulative settlements (cash payments and offsets). Using this measure, we define the timeline of firm level refund delays as follows:

- *Entry quarter* q_i^0 : the calendar quarter in which the firm’s cumulative approved balance first turns positive.
- *Exit quarter* q_i^1 : the first quarter in which at least 95% of the outstanding balance has been settled.
- *Delay* $D_i = q_i^1 - q_i^0$ (in quarters)⁷

This definition aggregates across individual claims and captures what really matters to firms: how long the government withholds money that is legally owed, and when the firm effectively regains access to it. By construction, only firms with a settled balance appear in the estimation sample. As Figure 2 shows, a large share of claims remained outstanding at the end of 2023. The estimation sample is therefore biased toward firms with shorter delays, making our estimates conservative with respect to the full population of claimants.

The leave-own-out instrument. To address the endogeneity concerns explained above, we instrument firm i ’s delay D_i with the leave-own-out mean delay of all other firms that entered the refund queue in the same quarter:

$$z_i = \frac{\sum_{j \neq i, q_j^0 = q_i^0} D_j}{|\{j \neq i : q_j^0 = q_i^0\}|}. \quad (9)$$

The key idea is that the instrument z_i captures the administrative congestion state of the ZRA at the time of entry, the backlog and staffing conditions faced by the cohort as a whole while excluding firm i ’s own delay from the average.

⁷We aggregate the data at the quarterly level because firms in Zambia correct VAT payments quarterly. According to ZRA officials, firms may file inconsistently monthly but will regularize these filings each quarter. Hence, when working with VAT data, a quarter aggregate is strongly suggested as it represents a true picture and reflects the actual filing behavior of firms.

Outcome construction. For each firm we define a pre-period (before q_i^0) and a during-period (from q_i^0 through q_i^1). The baseline outcome is the difference between the mean log outcome during the outstanding period and the mean log outcome before entry:

$$\Delta \ln y_i = \frac{1}{D_i} \sum_{q=q_i^0}^{q_i^1} \ln y_{iq} - \frac{1}{T_i^{\text{pre}}} \sum_{q < q_i^0} \ln y_{iq}, \quad (10)$$

where T_i^{pre} is the number of pre-entry observations. We estimate:

$$D_i = \alpha_1 + \eta z_i + \varepsilon_{i1}, \quad (11)$$

$$\Delta \ln y_i = \alpha_2 + \beta \hat{D}_i + \varepsilon_{i2}, \quad (12)$$

where $\hat{D}_i$ is the fitted value from the first stage. The coefficient now β identifies the causal effect of an additional quarter of delay on the log change in firm outcomes. One potential concern with the design in equation (10) is that the during-period window is D_i quarters long, so the mean outcome is implicitly averaged over more observations for firms with longer delays. If outcomes follow a trend within the delay window, the average will differ from what a fixed snapshot would show. As we show formally in Appendix B, the direction of the bias depends on the sign of the within-window trend: under the likely case of an underlying upward trend, the estimate understates the negative effect of delay, and should be interpreted as a lower bound.

To address this concern more directly, we complement the baseline with an fixed-window design in which the outcome is averaged over a fixed window of x quarters, not necessarily the delay window:

$$\Delta \ln y_i^{\text{ep}} = \frac{1}{x} \sum_{q \in [q_i^0 + x)}^{q_i^1} \ln y_{iq} - \frac{1}{T_i^{\text{pre}}} \sum_{q < q_i^0} \ln y_{iq}. \quad (13)$$

This outcome is not mechanically related to D_i through the window length but if the estimates using (10) and (13) are qualitatively similar, this provides reassurance that the baseline results are not driven by the mechanical correlation.

Exclusion restriction. The instrument must be relevant (correlated with D_i , verified directly in the first stage) and satisfy the exclusion restriction: z_i affects $\Delta \ln y_i$ only through its effect on D_i .

The principal threat to exclusion is a direct peer effect of other firms' delays on firm i 's outcomes (Angrist, 2014). In our setting, the delays of cohort peers

are determined by ZRA’s processing capacity, not by any behavior of firm i . It is difficult to conceive of a mechanism through which the duration of a competitor’s refund claim directly depresses firm i ’s sales or purchases, other than through the aggregate congestion that also delays firm i ’s own refund.

Another concern is that firms entering simultaneously may share common macroeconomic or industry conditions that independently affect outcomes. We dedicate Appendix C to this concern and show that the main IV results are difficult to reconcile with systematic pre-entry differential trends or with a simple adverse aggregate-shock explanation for high-congestion cohorts. In particular, show that the instrument does not affect outcomes in the periods leading up to the first claim by running event-study models as well as rerunning the analysis using only pre-claiming outcomes. By comparing firms which never claim for refunds with those in our main sample, we further counter concerns that high-delay quarters simply coincide with adverse aggregate economic conditions. In our main estimations, we aim to address these issue by adding fixed-effects for sector, location, and control for pre-refund firm size, such that our estimates rely on variation within industry-time-location cells. To further account for correlation within entry-quarters, while accounting for their relatively small number (34) at the same time, we rely on wild-bootstrap clustered standard errors for inference.

Heterogeneity by refund intensity. As shown in Section 4, the output distortion from delay depends not only on delay length D_i but also on the firm’s tax wedge $\tau_I - \tau_O$. To test this prediction empirically, we construct an observable proxy for this wedge. In particular, we define firm i ’s refund intensity by

$$S_i \equiv \frac{R_i^0}{\bar{P}_{i,\text{pre}}}, \quad (14)$$

where R_i^0 is the initial approved refund at the entry quarter and $\bar{P}_{i,\text{pre}}$ denote average quarterly purchases in the pre-entry period. A higher S_i means a larger fraction of working capital is withheld per quarter of delay implying a larger effective distortion.⁸ We have shown in our conceptual framework that conditional on the same delay, firms with higher S_i should suffer larger output losses. We test this empirically by splitting the sample into terciles of S_i and running the base-

⁸For a zero-rated exporter with effective output tax rate $\tau_O = 0$, the approved refund per quarter equals $\bar{P}_{i,\text{pre}} \tau_I$, so $S_i = \tau_I = \tau_I - \tau_O$ exactly. For firms with mixed outputs, S_i approximates $\tau_I - \tau_O$ with an error that depends on the share of standard-rated sales; we treat it as an empirical proxy for the structural wedge rather than an exact measure.

line IV separately within each group. This split-sample approach requires the same exclusion restriction as the baseline.⁹

6 Empirical Results

In this section, we present our empirical findings. We begin by discussing the effects of refund delays on sales and purchases, reported in quarterly VAT data. Then we cross check our results using yearly CIT data, a distinct data source, from which we also estimate the effects on outcomes such as profits, capital and employment.

6.1 Main results

We present our main findings in Table 1, which reports the estimated coefficients from our IV model in Eq. (12). In panel A, the outcome of interest is the difference between average sales during the delay window and average sales before a firm claimed for a refund (see Eq. (9)), while in panel B it is the equivalent measure for purchases. The first column presents the naive OLS estimate, which is close to zero and insignificant. As argued above, this attenuated effect could stem from endogenous selection of large firms into long delay episodes. In contrast, columns (2) to (5) report the $\hat{\beta}$'s estimated using the leave-own-out instrument and including different sets of controls. Importantly, the first stage F-statistic is large across specifications, confirming the validity of the instrument. All point estimates are negative, of similar magnitude and statistically significant based on the wild-cluster-bootstrap p-values. The estimates in panel A suggest that delaying the time between the first refund claim was made and the time the refund was settled to at least 95% by one additional quarter reduces a firm's sales by about 4.5 – 5.4 % relative to the time before refunds were claimed. Our conceptual framework predicts that such a reduction in output could be the result of frictions in working capital such as input purchases. We confirm this channel in panel B, where we observe a similarly negative and significant effect on purchases which is consistent across specifications.

In Appendix Table A.2, we report results when including sector-by-entry-year fixed effects. Together with the specification controlling explicitly for pre-claim

⁹A natural alternative would be to instrument the composite $D_i \times S_i$ directly, using $z_i \times S_i$ as a second instrument. In practice, this is not feasible: z_i has essentially no first-stage relevance for $D_i \times S_i$ ($F \approx 0.7$), because z_i shifts delay length but not refund intensity. The interaction $z_i \times S_i$ is near-proportional to S_i and therefore fails the exclusion restriction, as S_i is a strong positive predictor of outcomes in OLS ($\hat{\beta} \approx +0.21^{***}$ for sales), reflecting selection of high-intensity firms into the claimant pool.

Table 1: Effect of refund delays on sales and purchases

	(1) OLS	(2) IV	(3) IV	(4) IV	(5) IV
<i>Panel A: $\Delta \ln(\text{total sales})$</i>					
Delay (quarters)	0.002 (0.005)	-0.051*** (0.014)	-0.051*** (0.015)	-0.054*** (0.015)	-0.045*** (0.011)
First-stage F		122.1	114.3	116.4	112.3
Bootstrap p		0.001	0.002	0.001	0.000
Observations	1,829	1,829	1,828	1,791	1,712
<i>Panel B: $\Delta \ln(\text{total purchases})$</i>					
Delay (quarters)	-0.000 (0.005)	-0.042** (0.016)	-0.040** (0.016)	-0.043** (0.016)	-0.050*** (0.012)
First-stage F		152.2	125.7	127.8	112.5
Bootstrap p		0.024	0.034	0.020	0.000
Observations	1,848	1,848	1,848	1,810	1,708
Fixed effects	None	None	Sector	Province	Sect + Prov
Pre-entry size & purchases					✓

Notes: This table reports the estimated effect of refund delays on firm outcomes measured from VAT data. It reports the $\hat{\beta}$ coefficient as estimated by Eq. (12) (first row), the F-statistic of the first stage in Eq. (11) and the p-value from the wild cluster bootstrap (third row). Column (1) reports the naive OLS estimate. Columns (2)-(5) show results using the leave-own-out mean of other firms claiming in the same quarter as an instrument and consecutively add fixed effects and pre-claiming firm characteristics as controls. The outcome is the log change between mean outcome during the delay window and mean pre-entry outcome for both sales and purchases in Panel A and B, respectively. The sample is restricted to firms claiming after 2014 with delays of at least 1 quarter and whose refunds have been settled by 2023. Stars refer to the bootstrapped p-value: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

firm size, this provides one check against sector-specific economic conditions driving both cohort delays and firm outcomes. The point estimates remain negative and of similar order of magnitude, although they become less precise as the fixed effects absorb a substantial share of the variation in the cohort instrument. We therefore complement these specifications with direct pre-treatment diagnostics below.

While these results highlight how refund delays impact firms operational activities, we now turn to measure the effects on longer term measures such profitability, fixed assets and employment. To do so, we rely on data on firms' yearly CIT filings. Table 2 reports the results, with panel A showing the IV results including the full set of controls. As for the VAT outcomes, we observe no effect in the OLS specification for any of the outcomes and the F-statistic confirms validity of the instrument in all other specifications. Column (1) measures the effect of refund delays on sales, making use of a rare opportunity to cross-check our previous results using a distinct

Table 2: Effect of refund delays on profits, capital and employment

	(1) Sales	(2) Gross profit	(3) Fixed assets	(4) Employment
<i>Panel A: IV</i>				
Delay (quarters)	−0.038*** (0.012)	−0.037** (0.013)	−0.058*** (0.015)	−0.026* (0.012)
First-stage F	99.3	86.4	95.9	84.0
Bootstrap p	0.007	0.015	0.003	0.083
Observations	1,149	1,112	1,148	1,053
Fixed effects	Sector + Province			
Pre-entry controls	✓			
<i>Panel B: OLS</i>				
Delay (quarters)	0.003 (0.006)	0.003 (0.006)	0.000 (0.008)	0.002 (0.006)
Observations	1,224	1,181	1,314	1,149

Notes: This table reports the estimated effect of refund delays on firm outcomes measured from CIT data. Panel A reports the $\hat{\beta}$ coefficient as estimated by Eq. (12) (first row), the F-statistic of the first stage in Eq. (11) (second row) and the p-value from the wild cluster bootstrap (third row). The columns show results for different firm outcomes using the leave-own-out mean of other firms claiming in the same quarter as an instrument and consecutively add fixed effects and pre-claiming firm characteristics as controls. Panel B reports the naive OLS estimates. The outcome is the log change between mean outcome during the delay window (aggregated to years) and mean pre-entry outcome. The sample is restricted to firms claiming after 2014 with delays of at least 1 quarter and whose refunds have been settled by 2023. Stars refer to the bootstrapped p-value: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

and independent data source. Reassuringly, we estimate a strongly significant negative effect. The point estimate is slightly smaller than those for sales in Table 1. This is to be expected, however, because the delay period (over which the outcome is averaged; see Eq. (10)) is artificially larger when using CIT data. As the latter is reported yearly but delay windows may end within the year, the outcome is averaged over all years in which a refund has been outstanding. For example, even though the refund was outstanding for 1.5 years, the CIT based estimate will take the 2 year average. Hence, the estimates will be biased downward.

Column (2) shows the results for profits which imply that increasing delay by one additional quarter reduces yearly profits by 3.7% on average relative to the pre-claiming period. Columns (3) and (4) show how inputs, namely capital in the form of fixed assets and employment is affected. Both outcomes are affected negatively by refund delays. Firms reduce their fixed assets by 5.8% and even employment by 2.6%, suggesting severe input constraints during long waits for refund settlements.

Panel B shows that the OLS estimate is close to zero and insignificant, highlighting the endogeneity we address via using the instrument. These results imply that beyond harming operations such as sales and purchases, which occur more frequently, refund delays have severe consequences for longer-term measures of firm health and even has consequences for workers.

6.2 Pre-treatment diagnostics and identification validation

A central concern for the exclusion restriction is that firms entering the refund system in high-delay quarters may have been exposed to adverse economic conditions that independently affected their subsequent performance. We examine this possibility using three complementary diagnostics, described in the following. For specifications and more details, we refer the reader to Appendix C.

Pre-claiming event-study. First, we estimate event-time IV specifications for sales and purchases, normalizing outcomes to the quarter immediately preceding the first refund claim ($k = -1$). The pre-claim coefficients show no systematic movement as firms approach entry. Panel A of Appendix Table C.1 shows that a joint test of the seven coefficients from $k = -8$ to $k = -2$ fails to reject the null that they are jointly zero for sales as well as purchases (wild-cluster-bootstrap $p = 0.834$ and 0.695 , respectively). We also test for precision-weighted linear-trends and find slopes close to zero for both sales (0.0003 , $p = 0.898$) and purchases (0.0001 , $p = 0.959$). Thus, there is no evidence of a systematic pre-claim trend in the IV coefficients. Appendix Figure C.1 shows the full dynamic estimates and confidence intervals.

Pre-claiming placebo. Second, and complementary to the event-study, we construct a placebo outcome using only observations prior to the first refund claim. For each firm, we compare average log outcomes in quarters -4 to -1 with average log outcomes in quarters -8 to -5 and apply our main estimation model from Eq. (12). The estimated placebo effects for sales and purchases are close to zero and statistically insignificant (wild-cluster-bootstrap $p = 0.39$ and 0.78 , respectively; see panel B of Appendix Table C.1).

Comparison to non-claiming firms. Third, we directly examine the concern that high-delay entry quarters may simply coincide with adverse aggregate economic conditions. As these would arguably affect both firms who claim for refunds and

those who do not alike, we construct a sample of VAT filers that never claim for refunds. For each quarter and firm we calculate the residualized four-quarter sales and purchase growth and relate this to refund-cohort congestion in the same calendar quarter. The results are shown in Appendix Table C.2 In the raw time series, congestion is slightly negatively associated with non-claiming firms' sales and purchases growth. This relationship disappears completely, however, once a linear time trend is included: the coefficients become essentially zero and insignificant ($p = 0.975$ and 0.843 , respectively).¹⁰ The exercise therefore indicates that the negative raw correlation among non-claiming firms is a low-frequency time-series pattern rather than a persistent negative relationship between quarter-specific refund congestion and economy-wide firm performance. Applying the same strategy to our main estimation sample of firms claiming for a refund and the IV strategy, the negative significant estimates hold up both in the raw time series as well as when adding time trends.¹¹

Taken together, these three pieces of evidence speak in favor of the exclusion restriction to hold. Claimants do not exhibit systematic differences in trajectories before entering the refund system, an entirely pre-claim placebo outcome is economically small and statistically indistinguishable from zero, and there is no association between congestion and outcomes among non-claiming firms after accounting for secular calendar trends, while the claimant IV point estimates remain negative under linear, quadratic, and entry-year adjustments. These patterns are consistent with the leave-own-out instrument capturing variation in refund-processing conditions rather than simply proxying for adverse economic conditions at entry.

6.3 Channels of adjustment

We now turn to investigate the channels through which the negative headline result work. First, we note that our main results cannot be explained by specific sectors. As already confirmed by controlling for sectors, Appendix Table A.3 shows that the

¹⁰More flexible adjustments do not reveal a robust negative association either. With a quadratic trend the coefficients are positive but statistically insignificant, while year fixed effects produce positive estimates. Likewise, when congestion is residualized with respect to year fixed effects and a linear trend, the coefficients are positive (0.0110 for sales and 0.0129 for purchases) and their 95% confidence intervals include zero.

¹¹Although precision declines as increasingly flexible time controls absorb identifying variation, the point estimates remain negative and economically similar in magnitude, and the corresponding Kleibergen–Paap first-stage statistics remain between 25.1 and 106.0. As a deliberately demanding diagnostic, we also use only the residualized component of congestion as the excluded instrument. The estimates attenuate to -0.0137 for sales and -0.0287 for purchases, while the first-stage statistics fall to 9.47 and 9.31.

effects hold up when excluding any of the 21 sectors. Second, we test whether size of the refund matters, that is, firms with large outstanding claims relative to their input base experience more severe constraints when refund settlement is delayed. This is the prediction of Eq. (8) in our theoretical framework. We empirically distinguish different refund intensity by calculating Eq. (14), which gives the share of outstanding refunds in pre-claiming inputs and separate the sample along its distribution.

The results are shown in Table 3 and strongly support the theoretical prediction. Panel A shows that, while the low and middle tercile show a negative effect of delays on sales (-0.019 ; -0.03) the coefficients are statistically insignificant. The effect for the highest tercile, however, is highly significant and substantially larger than for the unrestricted sample (column (1)). In panel B, the effects on purchases reveal a similar pattern with the effect in the lowest tercile still significant at the 10% level, however. These results confirm the qualitative prediction of the model in Section 4: the delay distortion is proportional to the firm’s effective tax wedge $\tau_I - \tau_O$. Firms with large refund claims relative to their input base bear the bulk of the cost.

Following our theoretical framework, a large refund intensity is more likely when the input tax is small and thus, should be particularly relevant for firms dealing with exempt or zero-rated products. These are often exporters and, while the data does not allow us to directly distinguish exporters from non-exporters, we can leverage the VAT data to test which type of outputs and inputs are most affected. Table 4 reports these results for outputs in panel A and outputs in panel B. In both panels, we transform variables with many zeros using the inverse-hyberbolic-sign: $\ln(x + \sqrt{x^2 + 1})$. Confirming our hypothesis, we observe the largest decrease for zero-rated outputs of nearly 10%, which indeed points to a reduction in export activity due to delayed refunds. Further, we observe no effect for business to government sales (fourth row) which may imply that having outstanding refunds does not impede firms from securing government contracts. For inputs, capital expenditure decreases as much as standard rated purchases, suggesting lower investments due to refunds and thereby explaining the negative effect on fixed assets, which we document in Table 2. The strongest effects on inputs realizes for total taxable inputs, suggesting that firms avoid inputs on which VAT is due while awaiting outstanding refunds. Finally, we observe that refund delays exert a negative effect on total VAT due, which implies that less revenue is collected from VAT in total.

Taken together, these disaggregated results are consistent with the working-capital constraint mechanism: firms cut both production and investment in response to the liquidity withdrawal created by delayed refunds.

Table 3: Effect of refund delays on sales and purchases by refund intensity

	(1) Full sample	(2) Low S_i	(3) Mid S_i	(4) High S_i
<i>Panel A: $\Delta \ln(\text{total sales})$</i>				
Delay $D_i(\text{quarters})$	-0.045*** (0.011)	-0.030 (0.025)	-0.019 (0.015)	-0.060*** (0.020)
Bootstrap p	0.001	0.325	0.323	0.006
First-stage F	112.4	13.8	129.5	141.5
Observations	1,711	587	581	537
<i>Panel B: $\Delta \ln(\text{total purchases})$</i>				
Delay $D_i(\text{quarters})$	-0.050*** (0.012)	-0.050* (0.026)	0.001 (0.019)	-0.075*** (0.015)
Bootstrap p	0.000	0.084	0.898	0.000
First-stage F	112.6	14.3	108.6	162.9
Observations	1,707	590	581	530
Fixed effects	Sector $\times$ Province			
Pre-entry controls	✓			

Notes: This table reports the estimated effect of refund delays on firm outcomes measured from VAT data. It reports the $\hat{\beta}$ coefficient as estimated by Eq. (12) (first row), the F-statistic of the first stage in Eq. (11) and the p-value from the wild cluster bootstrap (third row). The columns show results using the leave-own-out mean of other firms claiming in the same quarter as an instrument. The outcome is the log change between mean outcome during the delay window and mean pre-entry outcome for both sales and purchases in Panel A and B, respectively. The different columns refer to firms in different terciles of the distribution of S_i , i.e. the relative size of the outstanding refund in pre-claiming inputs (defined in Eq. (14)). The sample is restricted to firms claiming after 2014 with delays of at least 1 quarter and whose refunds have been settled by 2023. All specifications include Sector $\times$ Province fixed effects and pre-claiming firm sales and purchases as controls. Stars refer to the bootstrapped p-value: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.4 Robustness

In this section, we test robustness of our main results by (i) overcoming the induced bias through the averaging over delay time and (ii) by testing the effects of refund delays on firms whose claims have not been settled by the end of our study period.

Fixed-window analysis. Finally, we assess to which extent our results are driven by the fact that the delay window is partly endogenous to the outcome measure. As discussed in Section 5 and shown in Appendix B, we estimate a lower bound because the outcome mechanically decreases with a longer delay window. We there-

Table 4: Effect of Refund Delays on Disaggregated VAT Outcomes

Outcome	Transf.	IV coef.	Std. err.	F	Bootsr. p	Obs.
<i>Output variables</i>						
Standard-rated outputs	Log	−0.024*	(0.012)	113.4	0.069	1,675
Zero-rated outputs	IHS	−0.093*	(0.048)	113.7	0.092	1,745
Taxable sales	Log	−0.042***	(0.012)	131.2	0.003	1,705
B&G sales	Log	0.005	(0.022)	85.8	0.832	1,623
<i>Input variables</i>						
Standard-rated purchases	Log	−0.034**	(0.012)	105.6	0.018	1,651
Capital expenditure	IHS	−0.034**	(0.015)	113.7	0.029	1,745
Total taxable inputs	IHS	−0.276*	(0.126)	113.7	0.066	1,745
Imports	IHS	−0.031	(0.031)	113.7	0.386	1,745
<i>Balance</i>						
Net VAT due	IHS	−0.080**	(0.029)	118.4	0.011	1,569
Fixed effects	Sector $\times$ Province					
Pre-entry controls	✓					

Notes: This table reports the estimated effect of refund delays on firm outcomes measured from VAT data. It reports the $\hat{\beta}$ coefficient as estimated by Eq. (12), the F-statistic of the first stage in Eq. (11) and the p-value from the wild cluster bootstrap.. All specifications include Sector $\times$ Province fixed effects and pre-claiming firm sales and purchases as controls. The outcome is the log change between mean outcome during the delay window and mean pre-entry outcome for the outcomes specified in each row. The sample is restricted to firms claiming after 2014 with delays of at least 1 quarter and whose refunds have been settled by 2023. Stars refer to the bootstrapped p-value: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

fore estimate the effect of refund delays on an outcome, which is independent of the actual delay. In particular, we define fixed waiting windows over which the outcome is averaged (see Eq. (13)) and re-estimate our baseline specification. The results are presented in Table 5 and clearly show that (i) the results are consistent and (ii) in line with our theoretical framework longer delays induce larger negative effects on both sales and purchases.

Including unresolved refund spells. One caveat to our strategy of defining delays is that we restrict the sample to firms whose outstanding refunds have been settled to at least 95%. Thus, there remains a sample of firms that face refund delays but are not included in our estimation. Appendix Figure A.2 plots the share of firms which have been settled by claiming cohort. In total, 7.15% of firms do not reach the threshold of 95% settlement by the end of 2023. To counter the concern that the exclusion of these firms is essential to our results, we estimate the effects for the excluded sample as follows.

Table 5: Robustness: Fixed-Window Outcome Designs

	(1)	(2)	(3)	(4)
	Mean-during	h=4 quarters	h=8 quarters	h=12 quarters
<i>Panel A: Sales</i>				
Delay (quarters)	-0.045*** (0.011)	-0.052*** (0.010)	-0.053*** (0.009)	-0.056*** (0.009)
Bootstrap p	0.001	0.000	0.000	0.000
First-stage F	112.4	109.4	117.6	119.1
Observations	1,711	1,694	1,709	1,716
<i>Panel B: Purchases</i>				
Delay (quarters)	-0.050*** (0.012)	-0.046*** (0.012)	-0.049*** (0.010)	-0.053*** (0.010)
Bootstrap p	0.000	0.001	0.000	0.000
First-stage F	112.6	110.1	112.1	112.1
Observations	1,707	1,683	1,694	1,703
Fixed effects	Sector $\times$ Province			
Pre-entry controls	✓			

Notes: This table reports the estimated effect of refund delays on firm outcomes measured from VAT data. It reports the $\hat{\beta}$ coefficient as estimated by Eq. (12), the F-statistic of the first stage in Eq. (11) and the p-value from the wild cluster bootstrap. All specifications include Sector $\times$ Province fixed effects and pre-claiming firm sales and purchases as controls. The outcome is the log change between mean outcome during a fixed window of quarters (see Eq. (13)) and mean pre-entry outcome for sales and purchases, respectively. The sample is restricted to firms claiming after 2014 with delays of at least 1 quarter and whose refunds have been settled by 2023. Stars refer to the bootstrapped p-value: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table A.4 progressively restricts the sample sample to earlier entry cohorts, which have substantially longer potential follow-up. As both the number of entry-quarter clusters and first-stage variation decline, precision falls in the earlier-cohort samples. Nevertheless, we observe estimates of similar magnitude to our baseline throughout. This suggests that the negative baseline estimates are not solely an artifact of excluding unresolved spells.

7 Conclusion

VAT is a key part of nearly every modern tax system and developing countries have been urged to implement VAT due to its various theoretical benefits. In this paper, we show that the low administrative capacity prevalent in many developing countries can impede the unfolding of these theoretical advantages and may even harm firms. Constructing a novel dataset, we show that in Zambia, refund payments

occur on average 700 days after they are claimed, whereas they are legally supposed to be paid out within 30 days. We identify the causal effect of refund delays on firm performance by constructing an instrument which uses the average delay of all other firms claiming for refunds within the same quarter. This measure reflects administrative congestion at the time of filing, is relevant and arguably satisfies the exclusion restriction. We merge the refund data to the universe of firms' tax filings, and estimate that delays have significant real effects on firm performance in terms of VAT (sales, purchases and VAT due) and CIT (gross profits, fixed assets and employment) outcomes.

Several pieces of evidence support the notion that these estimates reflect slow refund-processing rather than pre-existing firm trajectories or adverse aggregate conditions. Event-study estimates show no systematic differential movement before firms enter the refund system, the instrument is uninformative for firm outcomes before they claim refunds and there is no robust relationship between refund congestion and the performance of firms which never claim refunds. While these checks can not mechanically establish the exclusion restriction, together they substantially narrow the set of alternative common – shock explanations for the results.

We interpret our findings as a clear indication that observers' concerns about the negative consequences of slow refund processing are borne out in the data. Our findings confirm that observers' concerns regarding delayed refund processing are borne out by the data. Furthermore, these results serve as a caution that tax compliance measures, such as expanded withholding, cannot be evaluated in isolation from administrative capacity. Policies that expand refund eligibility or strengthen VAT enforcement inevitably increase the workload of refund administration. Unless tax administrations simultaneously invest in the capacity to process refund claims, reforms intended to improve compliance may unintentionally weaken firm performance, reduce trust in the tax system and ultimately undermine the tax base they seek to protect. The functioning of refund administration should therefore be viewed as a central component of effective tax policy rather than a purely administrative afterthought. Overall, our findings underscore the notion that reducing refund processing times should be a priority for tax administrations in developing countries. It is essential not only for preserving the neutrality of the VAT but also for supporting firm liquidity, investment and long-run economic growth.

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Appendix

A Additional Figures and Tables

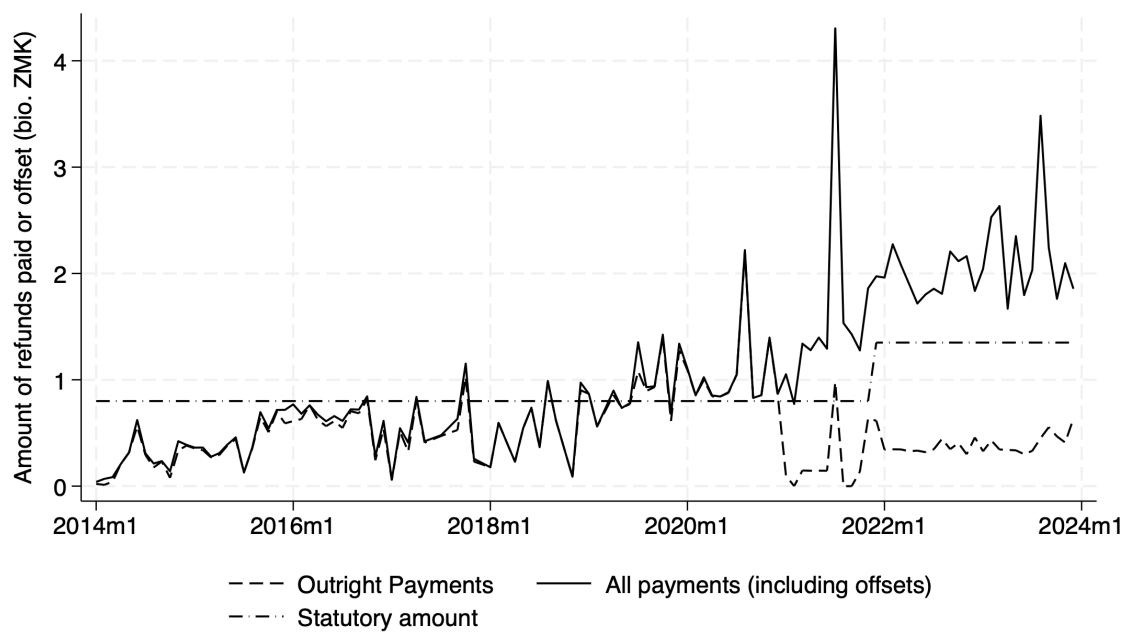
This appendix complements Section 3 by providing the overview of delays by sector (Table A.1) and the amount of offsets vs. outright payments (Figure A.1). It also complements Section 6 by showing the results including sector-by-entry-year fixed effects (Table A.2), excluding sectors (Table A.3), plotting the share of included firms by entry-cohort (Figure A.2) as well as showing the results for the excluded (not settled by 2023) firms for different entry-cohorts.

Table A.1: Refund payment delays in months

	<i>Mean</i>	<i>SD</i>	<i>90 pct.</i>	<i>95 pct.</i>
All firms	22.6	(21.2)	53	67
Mining	14.4	(13.7)	29	39
Transportation	21.5	(19.1)	49	63
Manufacturing	21.8	(19.1)	49	62
Construction	24.5	(20.8)	54	67
Agriculture	24.6	(25.7)	56	69
Wholesale & Retail	25.6	(22.5)	58	72

Notes: This table reports summary statistics for the delays in refund settling for different sectors. Refund delays are the difference in months between the 'normal' pay-out date i.e. 1 month and the actual payment month, as given by the payment files. Data source: ZRA. Years: 2014-2023.

Figure A.1: Offsets vs. Outright Payments as refund settlement



Notes: This figure plots the cumulative sum of VAT refund outright payments with and without offsets per year. The dotted line describes the official amount allocated to refund payments by ZRA. Data source: ZRA. Years: 2014–2023.

Table A.2: Main Results with Sector $\times$ Entry-Year Fixed Effects

	(1) Sales $\Delta \ln$	(2) Purchases $\Delta \ln$
Delay (quarters)	−0.034 (0.033)	−0.022 (0.032)
Bootstrap p	0.487	0.666
First-stage F	46.8	37.3
Observations	1, 814	1, 835
Fixed effects	Sector $\times$ entry year	

Notes: This table reports the estimated effect of re-fund delays on firm outcomes measured from VAT data. It reports the $\hat{\beta}$ coefficient as estimated by Eq. (12) (first row), the F-statistic of the first stage in Eq. (11) and the p-value from the wild cluster bootstrap (third row). The outcome is the log change between mean outcome during the delay window and mean pre-entry outcome for both sales and purchases. Both specifications include sector $\times$ year fixed effects. The sample is restricted to firms claiming after 2014 with delays of at least 1 quarter and whose refunds have been settled by 2023. Stars refer to the bootstrapped p-value: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

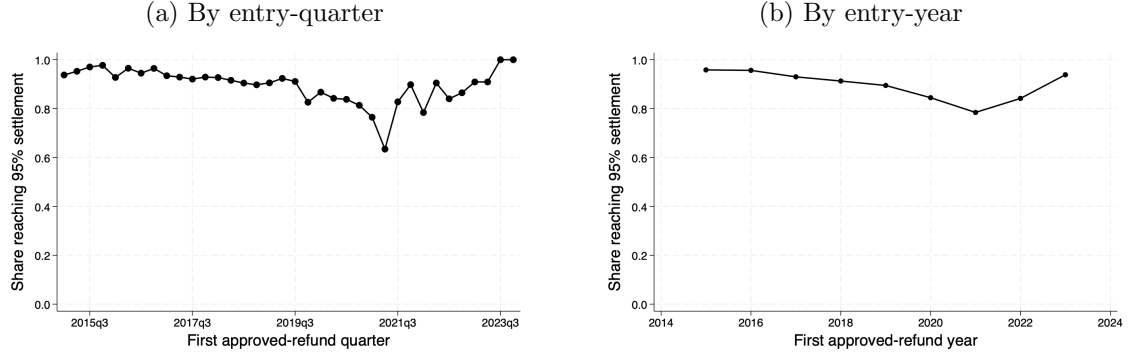
Table A.3: Leave-One-Out Sector Robustness

Sample	$\Delta \ln(\text{Sales})$			$\Delta \ln(\text{Purchases})$		
	Coef.	(SE)		Coef.	(SE)	
Full sample (baseline)	-0.047***	(0.011)	1,713	-0.053***	(0.012)	1,713
Drop: Accommodation	-0.048***	(0.011)	1,687	-0.054***	(0.012)	1,687
Drop: Households	-0.047***	(0.011)	1,712	-0.053***	(0.012)	1,712
Drop: Administration	-0.045***	(0.010)	1,631	-0.049***	(0.011)	1,631
Drop: Agriculture	-0.047***	(0.011)	1,658	-0.053***	(0.013)	1,658
Drop: Arts	-0.047***	(0.011)	1,706	-0.053***	(0.012)	1,706
Drop: Construction	-0.043***	(0.011)	1,575	-0.049***	(0.011)	1,575
Drop: Education	-0.046***	(0.011)	1,710	-0.052***	(0.012)	1,710
Drop: Electricity	-0.046***	(0.011)	1,702	-0.052***	(0.012)	1,702
Drop: Finance	-0.047***	(0.011)	1,706	-0.052***	(0.012)	1,706
Drop: Health	-0.047***	(0.011)	1,707	-0.053***	(0.012)	1,707
Drop: ICT	-0.049***	(0.011)	1,686	-0.053***	(0.012)	1,686
Drop: Manufacturing	-0.056***	(0.012)	1,544	-0.052***	(0.012)	1,544
Drop: Mining	-0.044***	(0.011)	1,670	-0.052***	(0.012)	1,670
Drop: Services	-0.044***	(0.011)	1,605	-0.051***	(0.013)	1,605
Drop: Professional	-0.048***	(0.011)	1,585	-0.053***	(0.013)	1,585
Drop: Government	-0.047***	(0.011)	1,712	-0.053***	(0.012)	1,712
Drop: Real estate	-0.047***	(0.011)	1,702	-0.053***	(0.012)	1,702
Drop: Transport	-0.045***	(0.012)	1,606	-0.055***	(0.014)	1,606
Drop: Utilities	-0.047***	(0.011)	1,700	-0.053***	(0.012)	1,700
Drop: Wholesale	-0.048**	(0.016)	943	-0.061***	(0.013)	943

Controls: province FE, pre-entry log sales, pre-entry log purchases.

Notes: This table reports the estimated effect of refund delays on firm outcomes measured from VAT data. Each row re-estimates the $\hat{\beta}$ coefficient as estimated by Eq. (12), i.e., baseline IV dropping all firms in the indicated sector. The outcome is the log change between mean outcome during the delay window and mean pre-entry outcome for both sales and purchases, respectively. The sample is restricted to firms claiming after 2014 with delays of at least 1 quarter and whose refunds have been settled by 2023. Stars refer to the bootstrapped p-value: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A.2: Settlement probability by refund-entry cohort



Notes: This figure plots the share of firms reaching 95% of cumulative approved refunds by the end of 2023. The left panel reports refund-entry-quarter cohorts and the right panel aggregates by refund-entry year.

Table A.4: Right-censoring diagnostics

Outcome	Sample / horizon	IV coefficient	Bootstrap p	F	N
Sales	Entry $\leq$ 2020q4	-0.0383	0.0408	42.81	1,647
	Entry $\leq$ 2019q4	-0.0353	0.3279	16.26	1,563
	Entry $\leq$ 2018q4	-0.0464	0.1932	11.64	1,219
Purchases	Entry $\leq$ 2020q4	-0.0373	0.0564	44.37	1,643
	Entry $\leq$ 2019q4	-0.0463	0.1684	16.79	1,554
	Entry $\leq$ 2018q4	-0.0567	0.1364	12.46	1,223

Notes: This table reports the estimated coefficients from estimating the baseline IV with firms who did not reach 95% of settlement after progressively restricting the sample to earlier refund-entry cohorts with longer potential follow-up. The outcome is the difference between the post- and pre-claiming period for sales and purchases, respectively. P-values are calculated from the wild cluster bootstrap with entry-quarter clustering and KP F is the Kleibergen–Paap first-stage statistic.

B Bias in the Mean-During Estimator

The outcome in Eq.(10) averages over a window of length D_i quarters. This introduces a potential bias when outcomes trend within the delay window. To see why, write the log outcome as

$$\ln y_{iq} = \ln y_{iq_i^0} + g \cdot (q - q_i^0) + u_{iq},$$

where g is a common within-window trend and u_{iq} is a mean-zero disturbance. The outcome is then:

$$\Delta \ln y_i^{\text{MD}} = (\ln y_{iq_i^0} - \bar{y}_i^{\text{pre}}) + g \cdot \bar{d}_i + \bar{u}_i,$$

where $\bar{d}_i = (D_i - 1)/2$ is the average distance from entry within the window. Instrumenting D_i with z_i :

$$\hat{\beta}^{\text{MD}} \approx \beta + \frac{g}{2} \frac{\text{Cov}(D_i, z_i)}{\text{Var}(D_i)}.$$

The bias term is $g/2 \times$ (a positive scalar). Under an upward trend ($g > 0$), the bias is positive, so $\hat{\beta}^{\text{MD}}$ is biased toward zero relative to the true effect $\beta < 0$. The mean-during estimate therefore provides a lower bound (in absolute value) on the true causal effect when firm outcomes trend upward within the delay window.

C Addressing the exclusion restriction

This appendix provides the estimating equations and results underlying the identification diagnostics summarized in Section 6.2. The exercises are intended to probe specific observable implications of the exclusion restriction rather than to establish it mechanically. In particular, they ask whether the cohort-level congestion instrument predicts outcomes before refund exposure, and whether the same congestion measure predicts adverse outcomes among firms which never claimed for refunds.

C.1 Pre-claiming event study.

Let q_i^0 denote firm i 's first approved-refund quarter and let $k = q - q_i^0$ be the time index relative to $q_i^0 - 1$. The outcome of interest at each the difference between the outcome y at relative time k and the outcome y at $q_i^0 - 1$:

$$\Delta \ln y_{ik} = \ln y_{i,q_i^0+k} - \ln y_{i,q_i^0-1}. \quad (15)$$

For each event time k , we estimate the IV system

$$D_i = \pi_k z_i + X_i' \delta_k + \nu_{ik}, \quad (16)$$

$$\Delta \ln y_{ik} = \alpha_k + \beta_k \hat{D}_i + X_i' \gamma_k + \varepsilon_{ik}, \quad (17)$$

where D_i is the total observed refund delay, z_i is the leave-own-out mean delay of other firms entering the refund system in the same quarter, and X_i contains the controls used in the corresponding specification.

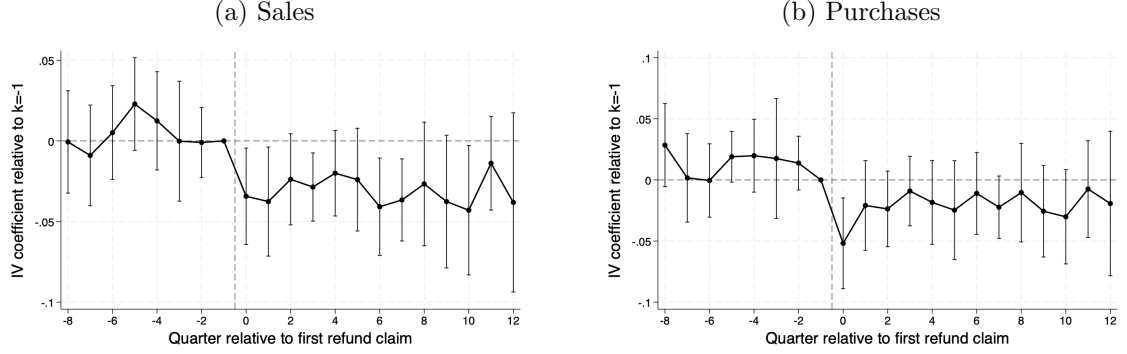
The coefficients for $k < 0$ should be interpreted as falsification coefficients rather than conventional difference-in-differences event-study treatment effects. They ask whether firms assigned longer subsequent delays by cohort congestion were already evolving differently before entering the refund system. The resulting estimates from Eq. (17) for each k are shown in Figure C.1 and panel A of Table C.1 tests the joint null

$$H_0 : \beta_{-8} = \beta_{-7} = \dots = \beta_{-2} = 0. \quad (18)$$

C.2 Pre-claiming placebo

As a second complementary diagnostic, we test whether the instrument predicts differential firm performance even before refund exposure, as could occur if entry-

Figure C.1: Event-time IV pre-treatment diagnostics



Notes: This figure plots the dynamic IV coefficients $\hat{\beta}_{ks}$ as estimated by Eq. (17) for sales (panel A) and purchases (panel B). Each coefficient estimates the effect of one additional quarter of instrumented total refund delay on the log outcome at event time k relative to $k = -1$. Bars denote 95% confidence intervals with standard errors clustered by refund-entry quarter.

Table C.1: Pre-treatment diagnostics for the leave-own-out instrument

	Sales	Purchases
Panel A: Joint event-time pre-trend tests, $k = -8, \dots, -2$		
Cluster-robust joint p -value	0.625	0.440
Bootstrap joint p	0.834	0.695
Linear pre-trend slope	0.0003	0.0001
Linear pre-trend p	0.898	0.959
Panel B: Pre-period placebo IV		
IV coefficient	-0.0097	0.0017
Clustered standard error	(0.0102)	(0.0098)
Bootstrap p	0.392	0.782
First-stage F	199.1	235.4
Observations	997	942

Notes: This table reports the results from jointly testing the $\hat{\beta}_{ks}$ as estimated in Eq. (17) in panel A. The joint tests assess the null that the seven pre-claim coefficients are jointly zero. Rows 2 and 3 show the estimated pre-trend slope and corresponding p -value. Panel B defines the placebo outcome as the difference between average log outcomes in quarters -4 to -1 and quarters -8 to -5 before the first refund claim. Refund delay is instrumented with the leave-own-out mean delay of other firms entering the refund system in the same quarter. Panel B includes sector and province fixed effects and controls for early pre-period sales and purchases. Inference is clustered by entry quarter using wild-cluster-bootstrap p -values.

quarter congestion were systematically related to pre-existing firm trajectories or economic conditions. To do so, we imitate our main estimating Eq. (12) but redefine

our outcome to exclusively rely on data prior to firms' first claiming:

$$\Delta \ln y_i^{\text{pre}} = \frac{1}{4} \sum_{k=-4}^{-1} \ln y_{i,q_i^0+k} - \frac{1}{4} \sum_{k=-8}^{-5} \ln y_{i,q_i^0+k}. \quad (19)$$

We then estimate

$$D_i = \pi z_i + X_i' \delta + \nu_i, \quad (20)$$

$$\Delta \ln y_i^{\text{pre}} = \alpha + \beta^{\text{pre}} \widehat{D}_i + X_i' \gamma + \varepsilon_i. \quad (21)$$

The results for sales and purchases are reported in panel B of Table C.1. We observe small and insignificant effects, suggesting that the instrument is not merely picking up pre-existing differential trends but is informative only for what happens after claiming for refunds.

C.3 Comparison to non-claiming firms

A separate threat is that high-congestion quarters coincide with adverse aggregate conditions that reduce firm performance independently of refund exposure. To examine this possibility, we use VAT filers which never claim for refunds and check whether congestion also matters for their performance.

Specification for non-claiming firms. Let g_{iq}^{NC} denote four-quarter outcome growth for non-claiming firm i in calendar quarter q , and let C_q denote mean delay of all firms claiming for refunds in quarter q . The diagnostic regression is

$$g_{iq}^{NC} = \alpha + \theta C_q + X_{iq}' \gamma + f(q) + u_{iq}. \quad (22)$$

where $f(q)$ is varied across specifications: a linear trend, a quadratic trend, and calendar-year fixed effects. For the regression, firm-level growth is first residualized with respect to the relevant controls. Clearly, Eq. (22) can not produce an estimate of the causal effect of refund delays, since these firms never face a refund delay. Instead, it asks whether the same congestion in a given quarter which provides identifying variation among claimants systematically predicts economic performance among other VAT filers.

Panel A of Table C.2 shows how the estimated relationship changes as calendar-time variation is removed. In column (1), the raw association is negative for both outcomes. Once a linear calendar trend is included, however, the coefficients move

Table C.2: Common-shock diagnostics and calendar-time adjustments

	Raw	Linear trend	Quadratic trend	Year FE	Residual congestion
<i>Panel A: Non-claiming firms: $\hat{\theta}$</i>					
Sales	-0.0072*** (0.0024)	0.0001 (0.0046)	0.0087 (0.0054)	0.0114** (0.0047)	0.0110* (0.0057)
Purchases	-0.0077*** (0.0026)	-0.0010 (0.0050)	0.0080 (0.0052)	0.0126*** (0.0043)	0.0129* (0.0068)
<i>Panel B: Claiming firms' IV estimates</i>					
Sales	-0.0462*** (0.0114)	-0.0392* (0.0177)	-0.0383 (0.0229)	-0.0349 (0.0268)	-0.0137 (0.0277)
KP F	105.87	63.13	25.07	34.60	9.47
Purchases	-0.0507*** (0.0120)	-0.0574** (0.0170)	-0.0379 (0.0230)	-0.0504 (0.0254)	-0.0287 (0.0287)
KP F	106.00	73.05	27.51	38.43	9.31

Notes: This table reports the coefficient $\hat{\theta}$ as estimated from Eq. (22) for non claiming firms in panel A. The outcome is the mean residualized four-quarter log growth. The linear and quadratic specifications add calendar-time trends, and the year-FE specification adds calendar-year fixed effects. Residual congestion is the component of congestion remaining after removing year fixed effects and a linear trend. Panel B reports IV estimates $\hat{\beta}$ as estimated from Eq. (24) for firms claiming refunds, instrumenting individual refund delay with the leave-one-out cohort delay except in the final column, where the excluded instrument is residual congestion. Standard errors clustered by entry quarter are in parentheses. Stars in Panel B are based on null-restricted wild-cluster-bootstrap p -values; the corresponding values are 0.0002, 0.0858, 0.2190, 0.3359, 0.6757 for sales and 0.0010, 0.0244, 0.2422, 0.1794, 0.3801 for purchases. Panel A stars use heteroskedasticity-robust conventional p -values.

essentially to zero. Quadratic trends, year fixed effects, and a measure of congestion residualized with respect to year fixed effects and a linear trend do not produce a robust negative association. The relevant conclusion is therefore not that congestion is unrelated to all aggregate conditions, but that the negative raw relationship that could rationalize the main IV estimates through an adverse common shock is not robust to accounting for low-frequency calendar-time movements.

Alternate specification for claiming firms. The above exercise indicates that the raw relationship between congestion and economic activity contains a low-frequency calendar-time component. We therefore impose analogous calendar-time adjustments directly on IV in our main (claiming firms) sample. That is, we estimate:

$$D_i = \mu + \pi z_i + X_i' \delta + f(q) + \nu_i, \quad (23)$$

$$\Delta \ln y_i = \alpha + \beta \hat{D}_i + X_i' \gamma + f(q) + \varepsilon_i, \quad (24)$$

where $f(q)$ is defined as above. Because the instrument D varies at the entry-cohort level, the time controls necessarily absorb part of its identifying variation. Thus, Eqs. (23) and (24) can not require statistical precision to remain unchanged as increasingly flexible time controls are introduced. Instead, the question is whether the economically negative relationship between instrumented delay and the outcome disappears once the low-frequency component identified in the non-claiming firms exercise is removed.

Panel B of Table C.2 reports these estimates. The negative point estimates remain of similar order of magnitude when adding linear and quadratic entry-quarter trends or entry-year fixed effects, although confidence intervals widen as these controls absorb variation in the cohort-level instrument.